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Total No. of printed pages = 02

Monsoon, 2024  
B. Tech 1<sup>st</sup> Semester Examination  
Programming for Problem Solving- Theory  
Course Code: BCS23101T

Full Marks - 50

Time - 2 hours

The figure in the margin indicates full marks for the questions.

Part A

Answer ALL questions  
(10 x 1 mark = 10 marks)

1. a) Which of the following valid identifiers in C?  
i) 123abc    ☒ ii) \_abc123    iii) @variable    iv) double
- b) Which of the following is a constant in C?  
i) 10    ii) float    iii) int x = 5;    ☒ iv) #define PI 3.14

- c) What is the output of the following code?

```
int i = 0;
for(; i < 5; i++) {
    if(i == 3) {
        break;
    }

    printf("%d ", i);
}
```

- i) 0 1 2 3    ii) 0 1 2    ☒ iii) 1 2 3    iv) 0 1

- d) Consider the following code. What is the value of x after execution?

```
int x = 0;
for(int i = 0; i < 5; i++) {
    if(i % 2 == 0) {
        x += i;
    } else {
        x -= i;
    }
}
```

- i) -5    ☒ ii) 5    iii) 0    iv) -10

- e) Which of the following is the correct way to declare a function in C?  
☒ i) `int func(void);`    ii) `int func[];`    iii) `int func();`    iv) `function int func();`
- f) What is the size of an integer array with 5 elements in C?  
 i) 5 bytes    ii) 20 bytes    iii) 4 bytes    ☒ iv) `5 * sizeof(int)` bytes
- g) What is the correct syntax for calling a function named `sum()` that takes two integers as parameters?  
 i) `sum(int a, int b);`    ☒ ii) `sum(a, b);`    iii) `sum(int, int);`    iv) `sum(a, b)`
- h) Which of the following is a correct definition of a structure in C?  
 i) `struct { int a, b; };`  
☒ ii) `struct point { int x; int y; };`  
 iii) `struct Point { int a = 1; };`  
 iv) `structure point { int x; };`
- i) What is the primary use of function?  
 i) Decision making  
 ii) Printing output  
☒ iii) Code reusability  
 iv) Variable declaration
- j) What character is used to terminate a C string?  
 i) n    ii) ;    ☒ iii) \0    iv) w

#### Part B

Answer ANY FOUR questions

(4 x 5mark = 20 marks) (Max.100 words each)

2. Write a program to find the sum of odd numbers between 1 to 100.
3. Write about two operators used in C language.
4. Write a program to find the largest number in an array.
5. Write about the compilation and execution process of a C program.
6. What is the difference between while loop and do while loop.

#### Part C

Answer ANY TWO questions

(2 x 10mark = 20 marks) (Max.200 words each)

7. What is the difference between array and structure. Write a program to read and display 100 employee records using array of structure.
8. Write a program to perform addition of two matrices. Check the necessary conditions.
9. Write C programs to convert Fahrenheit to celsius using
  - a) Function with return type and with arguments
  - b) Function with no return type and with arguments
10. Write notes on the following:
  - a) 1D array
  - b) String manipulation functions